**R1**

| sid | bid | day |
|---|---|---|
| 22 | 101 | 10/10/96 |
| 58 | 103 | 11/12/96 |

**B1**

| bid | bname | color |
|---|---|---|
| 101 | tiger | red |
| 103 | lion | green |
| 105 | hero | blue |

**S1**

| sid | sname | rating | age |
|---|---|---|---|
| 22 | dustin | 7 | 45.0 |
| 31 | lubber | 8 | 55.5 |
| 58 | rusty | 10 | 35.0 |

**S2**

| sid | sname | rating | age |
|---|---|---|---|
| 28 | yuppy | 9 | 35.0 |
| 31 | lubber | 8 | 55.5 |
| 44 | guppy | 5 | 35.0 |
| 58 | rusty | 10 | 35.0 |

**Q1:** $\pi_{sid,sname,age}(\sigma_{rating<10}(S_2))$

**Q2:** $\pi_{sname,rating}(\sigma_{S_1.sid>R_1.sid}(S_1 \times R_1))$

**Q3:** $\pi_{sid,bname}(\sigma_{bid}(R_1 \times B_1))$

**Q4:** $S_2/\left(\pi_{sid,age}(S_2)/\pi_{sid}(\sigma_{bid=103}(R_1))\right)$

**Q5:** $(S_2 \bowtie_* R_1) \underline{\cup} (B_1 *\bowtie_* R_1)$

**Q6:** $\pi_{sname,rating,age,bname,color}\left(((S_1 \cup S_2) \bowtie_{sid} R_1) \bowtie_{bid} \sigma_{color=red}(B_1)\right)$

**Q7:** $\pi_{sid}(S_2) - \pi_{sid}(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2))$

**Q8：Conver the query of Q1 to DRC formula and TRC formula.**

**Q9：Conver the query of Q2 to DRC formula and TRC formula.**

**Q10：Conver the query of Q3 to DRC formula and TRC formula.**

**Q11: What's data model? What's data schema?**

**Q12: Students(sid: string, name: string, ID number: string, cid: string)**

**Courses(cid: string, cname:string)**

**List all super keys, candidate keys, primary keys, foreign keys of table Students.**

**Q13: What are the relational operations on the relational data model? What are their differences and similarities?**